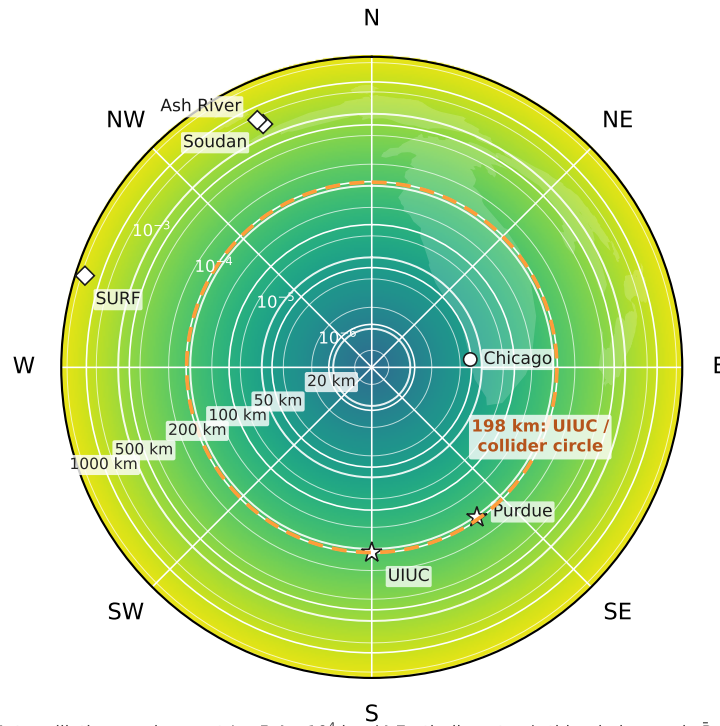


Tau appearance around the compass, stage by stage

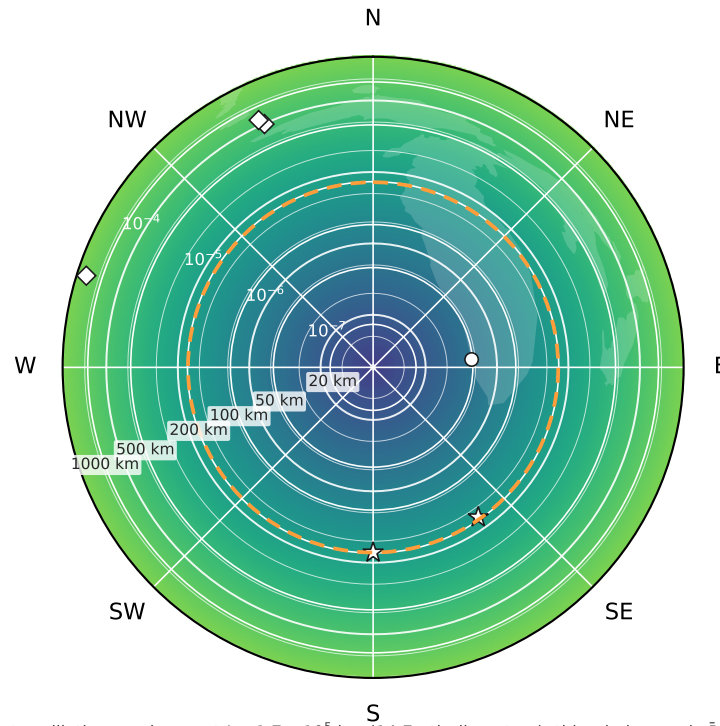
colour = fraction of the stage's $\bar{\nu}_\mu$ flux that has become $\bar{\nu}_\tau$ by the range where the beam surfaces at that bearing (it depends on range only: rings). White: lakes. Orange dashed: the 198 km circle the collider and the co-tilted chain exit on.

RCS1 63 → 314 GeV $\langle E_\nu \rangle_{axis} \approx 110$ GeV



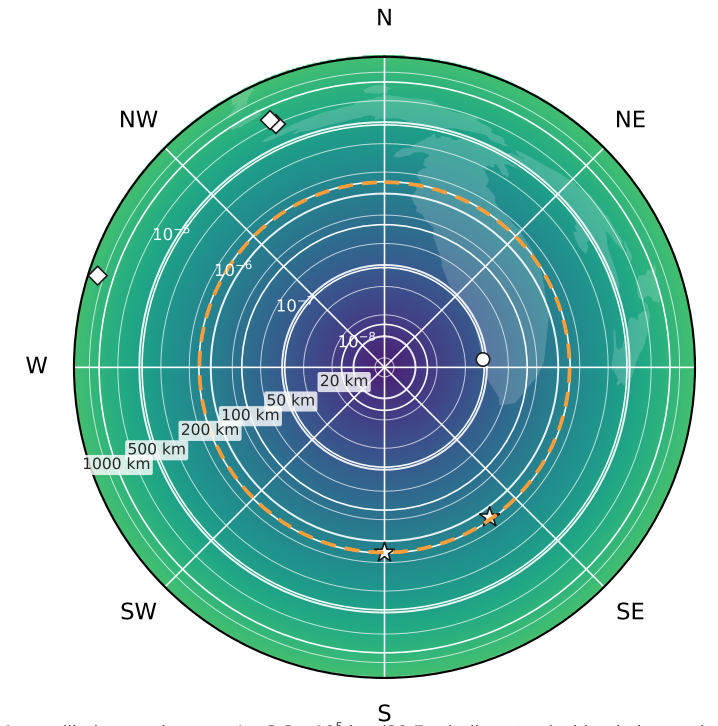
1st oscillation maximum at $L = 5.4 \times 10^4$ km (4 Earth diameters): this whole map is $\bar{P} \propto L^2$
 UIUC 198 km: $\bar{P} = 1.1 \times 10^{-4}$ · Soudan 736 km: 1.5×10^{-3}
 on-axis $\nu_\tau + \bar{\nu}_\tau$ CC: 0.0174 per tonne-yr, at any range

RCS2 314 → 750 GeV $\langle E_\nu \rangle_{axis} \approx 351$ GeV



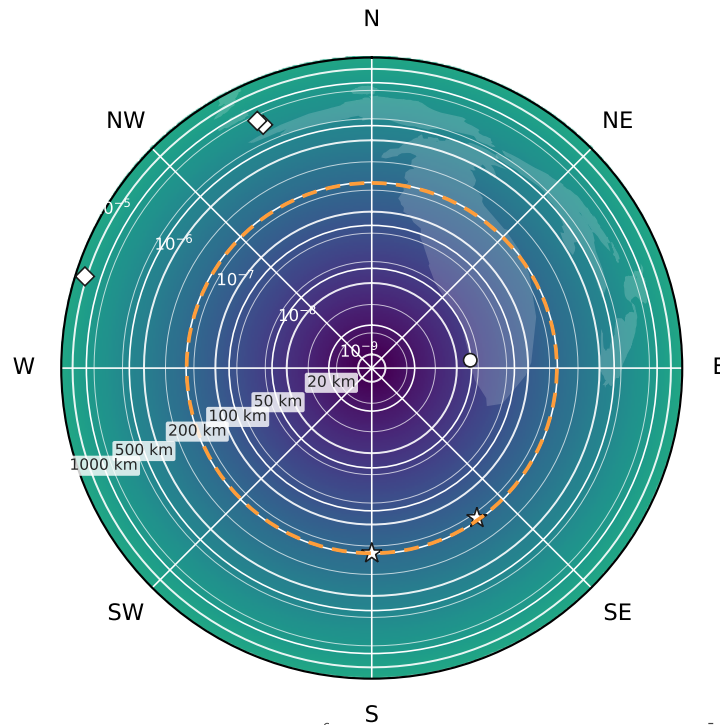
1st oscillation maximum at $L = 1.7 \times 10^5$ km (14 Earth diameters): this whole map is $\bar{P} \propto L^2$
 UIUC 198 km: $\bar{P} = 7.2 \times 10^{-6}$ · Soudan 736 km: 9.8×10^{-5}
 on-axis $\nu_\tau + \bar{\nu}_\tau$ CC: 0.0623 per tonne-yr, at any range

RCS3 750 → 1500 GeV $\langle E_\nu \rangle_{axis} \approx 758$ GeV



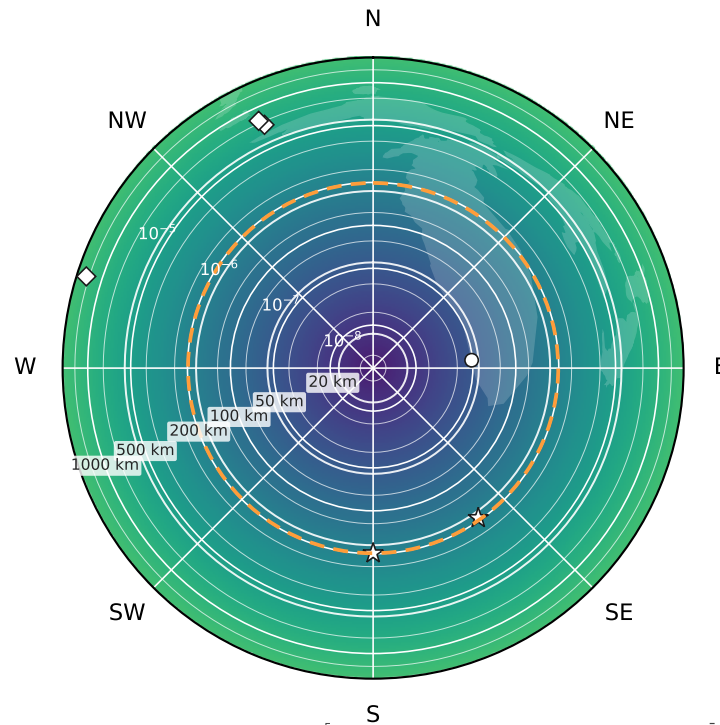
1st oscillation maximum at $L = 3.8 \times 10^5$ km (29 Earth diameters): this whole map is $\bar{P} \propto L^2$
 UIUC 198 km: $\bar{P} = 1.4 \times 10^{-6}$ · Soudan 736 km: 2.0×10^{-5}
 on-axis $\nu_\tau + \bar{\nu}_\tau$ CC: 0.0405 per tonne-yr, at any range

RCS4 1500 → 5000 GeV $\langle E_\nu \rangle_{axis} \approx 2036$ GeV



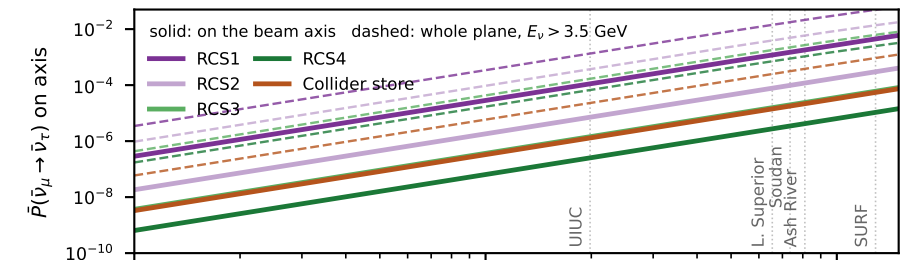
1st oscillation maximum at $L = 1.0 \times 10^6$ km (79 Earth diameters): this whole map is $\bar{P} \propto L^2$
 UIUC 198 km: $\bar{P} = 2.5 \times 10^{-7}$ · Soudan 736 km: 3.5×10^{-6}
 on-axis $\nu_\tau + \bar{\nu}_\tau$ CC: 0.108 per tonne-yr, at any range

Collider store 5000 GeV $\langle E_\nu \rangle_{axis} \approx 1806$ GeV

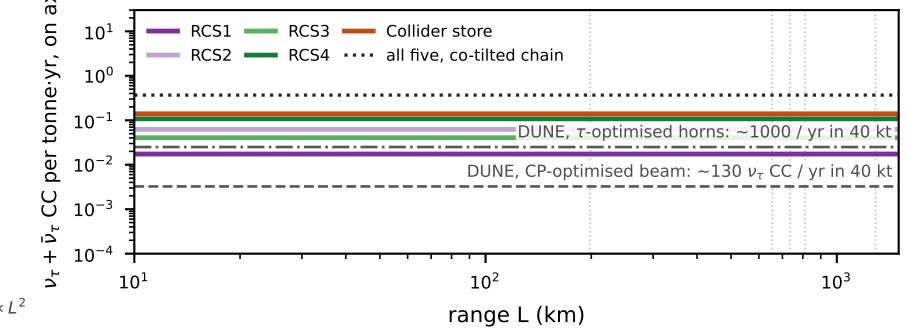


1st oscillation maximum at $L = 9.0 \times 10^5$ km (70 Earth diameters): this whole map is $\bar{P} \propto L^2$
 UIUC 198 km: $\bar{P} = 1.3 \times 10^{-6}$ · Soudan 736 km: 1.8×10^{-5}
 on-axis $\nu_\tau + \bar{\nu}_\tau$ CC: 0.14 per tonne-yr, at any range

the same curves, all stages



rate for a point detector on the axis: baseline-independent



$\bar{\nu}_\tau$ appearance fraction $\bar{P}$ of the stage's $\bar{\nu}_\mu$ flux, on the beam axis, at that range ($\Delta m^2 = 2.5 \times 10^{-3}$ eV², amplitude 0.95); contours at 1, 2, 5 $\times 10^k$

Point spectra: the RCS pencils are $1/\gamma$ -sharp, so a point on the axis sees the hard forward spectrum ($\langle E_\nu \rangle = 0.7 E_\mu$); the collider plume is smeared to $\sigma_\theta = 7/\gamma$, so its axis sees the whole-plume spectrum ($\langle E_\nu \rangle = 0.35 E_\mu$, cut at $E_\nu/E_\mu = 1/50$) at 1/101 of the pencil density. Rates: north-aimed decays per cycle from chain_timing.json, 5 Hz, 1.2×10^7 s/yr, both signs, CSMS CC cross-sections, τ -threshold suppression; on-axis flux $\propto 1/L^2$ and $\bar{P} \propto L^2$ cancel, so the rate is flat in L . DUNE lines: TDR vol. II §4.1.1.3 ν_τ CC interactions per year before detector efficiency (130 CP-optimised, ~ 1000 τ -optimised) in the 40 kt fiducial far detector at 1300 km, per tonne.